THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

AMTI – NMTC – **25th October, 2026** – SUB-JUNIOR FINAL

Timings: 10am to 1pm

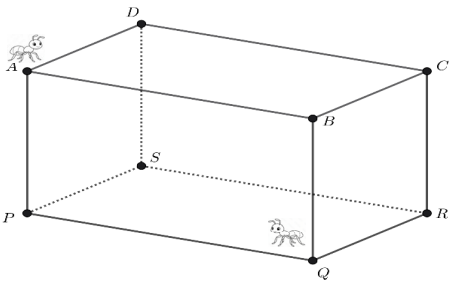
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Instructions:

1. There are 8 questions, answer all questions. Each question carries 10 marks.
2. Elegant and innovative solutions will get extra marks.
3. Diagram and justification should be given wherever necessary.
4. Before start answering, fill in the FACE SLIP completely.
5. Your **Rough Work** should be in the answer sheet itself.
6. Diagrams given here are only visual aids; they are NOT drawn to scale.

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1. A rectangular box is rested on a horizontal surface. Length, breadth and height of the box are 20*cm*, 12*cm* and 9*cm* respectively. The box has 5 faces; it is open from the top. Its vertices are named A, B, C, D, P, Q, R, S as shown in the figure. There are two ants; *ant*-1 at the vertex A and *ant*-2 at the vertex Q inside the box at rest. Neither ant can fly, but only can crawl along the face or edge of the box. *Ant*-1 has to reach at the vertex R going along the edges or a face of the box from outside of the box and *ant*-2 has to reach vertex D going from inside the box. Both ants start at the same time at the same speed of 1*cm/s*, and both follow the shortest possible route. Find the time taken by both the ants to reach their respective destinations.



2. In the regular pentagon ABCDE, the perpendicular at C to CD meets AB at F.
Prove that AE + AF = BE.

3. Lata wants to multiply a two-digit number with a three-digit number. When she simply placed the two-digit number to the left of the three-digit number, forming a five-digit number, it is found that this number is exactly nine times the product of the two numbers Lata took. Find the two numbers.

4. A grocer used a defective balance and gave 4.8*kg* of almond at Rs. 4800 per *kg* to a customer. The customer was in doubt about the weight. So, he asked the grocer to put the almond bag in the other pan. Surprisingly it weighed only 2.7*kg*. The grocer apologised and charged the amount for only 3750*g* (that is average of 4.8*kg* and 2.7*kg*) of almond. In doing so, the grocer still earned the profit. Find the percentage profit of the grocer.

5. Let x be a five-digit number. Let y be the four-digit number formed from x by deleting its middle digit. Find all such x when y divides x completely.

6. $A_1, A_2, A_3, \ldots, A_{20}$ are the vertices of a regular polygon of 20 sides. $A_1A_2A_3P_1$ is a square on A_1A_2 interior of the polygon. $A_3Q_3Q_4Q_5A_4$ is the regular pentagon drawn on side A_3A_4 interior of the polygon. Prove that Q_3 lies on the line joining A_3P_2.

7: Suppose $a_1, a_2, a_3, a_4 \ldots, a_n$ is a sequence of numbers with the property that

$$a_1 + a_2 + a_3 + \cdots + a_n = \frac{n(n+1)(n+2)}{6}, \text{for all natural numbers } n.$$

[1] Find a_{100}, [2] Evaluate: $\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \cdots + \frac{1}{a_{89}} + \frac{1}{a_{90}}$.

8: Let $a < b < c < d < e$ be real numbers. Among 10 sums of the pairs of these numbers, the smaller three are 32, 36, 37, while the larger two are 48 and 51. Find all possible values of e.